

Low Cycle Fatigue: Core Physics and Equations

Low Cycle Fatigue MCP Project

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Abstract

A brief reference for the physics and governing equations behind low cycle fatigue (LCF) analysis. This document defines the quantities the analysis tool computes. All analysis uses *true* stress and *true* strain.

1 Background

Fatigue is the progressive, localized damage a material accumulates under cyclic loading, leading to failure at stresses below the monotonic ultimate strength. *Low cycle fatigue* (LCF) refers to the regime in which each cycle imposes significant plastic deformation, so failure occurs in a relatively small number of cycles (typically 10^1 to 10^4). LCF is strain-controlled: a specimen is cycled between fixed strain limits, and the stress response is measured.

Throughout, we use true stress σ and true strain ε , related to engineering quantities by

$$\varepsilon = \ln(1 + \varepsilon_{\text{eng}}), \quad \sigma = \sigma_{\text{eng}} (1 + \varepsilon_{\text{eng}}). \quad (1)$$

2 Cyclic Stress-Strain Response

LCF tests are typically run under *fully reversed* strain control, characterized by the strain ratio

$$R = \frac{\varepsilon_{\min}}{\varepsilon_{\max}} = -1, \quad (2)$$

i.e. each cycle swings symmetrically between equal tensile and compressive strains. Under this loading the stress-strain path traces a closed *hysteresis loop* each cycle. The total strain amplitude decomposes into elastic and plastic parts:

$$\frac{\Delta\varepsilon_t}{2} = \frac{\Delta\varepsilon_e}{2} + \frac{\Delta\varepsilon_p}{2}, \quad \frac{\Delta\varepsilon_e}{2} = \frac{\Delta\sigma}{2E}, \quad (3)$$

where $\Delta\sigma$ is the stress range, E is Young's modulus, and the plastic amplitude $\Delta\varepsilon_p/2$ is measured from the width of the loop.

As cycling proceeds the material typically *hardens* (peak stress rises) and then *softens* (peak stress falls) toward failure. A tension/compression asymmetry in the peak stresses indicates direction-dependent deformation, quantified by the ratio

$$R_{TC} = \frac{|\sigma_{\max, \text{tension}}|}{|\sigma_{\max, \text{compression}}|}. \quad (4)$$

Note that even under fully reversed strain control ($R = -1$), this stress ratio need not equal unity. A value of $R_{TC} \neq 1$ reflects a nonzero mean stress in the cycle (see Morrow's correction, §??).

3 Cyclic Energy Density

The area enclosed by one hysteresis loop is the mechanical energy dissipated per unit volume per cycle:

$$W = \oint \sigma(\varepsilon) d\varepsilon \quad [\text{J m}^{-3}]. \quad (5)$$

It is reported at the peak-hardened state and at the half-life (stable) cycle, conventionally in MJ m^{-3} .

4 Strain-Life Relations

4.1 Total strain-life

Fatigue life is modeled as the sum of elastic (Basquin) and plastic (Coffin-Manson) contributions versus the reversals to failure $2N_f$:

$$\frac{\Delta\varepsilon_t}{2} = \underbrace{\frac{\sigma'_f}{E} (2N_f)^b}_{\text{elastic (Basquin)}} + \underbrace{\varepsilon'_f (2N_f)^c}_{\text{plastic (Coffin-Manson)}}. \quad (6)$$

4.2 Basquin (elastic branch)

$$\frac{\Delta\sigma}{2} = \sigma'_f (2N_f)^b, \quad (7)$$

with fatigue strength coefficient σ'_f and exponent b obtained from a linear fit on $\log(\Delta\sigma/2)$ versus $\log(2N_f)$.

4.3 Coffin-Manson (plastic branch)

$$\frac{\Delta\varepsilon_p}{2} = \varepsilon'_f (2N_f)^c, \quad (8)$$

with fatigue ductility coefficient ε'_f and exponent c obtained from a linear fit on $\log(\Delta\varepsilon_p/2)$ versus $\log(2N_f)$.

4.4 Morrow mean-stress correction

The base strain-life relation assumes a zero mean stress. When the cycle carries a nonzero mean stress σ_m (e.g. from the tension/compression asymmetry $R_{TC} \neq 1$, even at a strain ratio of $R = -1$), Morrow's relation corrects the elastic (Basquin) term by shifting the fatigue strength coefficient by σ_m :

$$\frac{\Delta\varepsilon_t}{2} = \frac{\sigma'_f - \sigma_m}{E} (2N_f)^b + \varepsilon'_f (2N_f)^c, \quad (9)$$

where the mean stress of a cycle is

$$\sigma_m = \frac{\sigma_{\max} + \sigma_{\min}}{2}. \quad (10)$$

A tensile mean stress ($\sigma_m > 0$) reduces fatigue life, while a compressive mean stress ($\sigma_m < 0$) extends it. The relation reduces to the base strain-life equation when $\sigma_m = 0$. Morrow's correction leaves the plastic (Coffin-Manson) term unchanged, on the assumption that mean stress relaxes in the high-plasticity regime and chiefly affects the elastic, longer-life contribution.

5 Cyclic Stress-Strain Curve (Ramberg-Osgood)

The stabilized stress-strain behavior across amplitudes is described by

$$\Delta\varepsilon_t = \frac{\Delta\sigma}{E} + \left(\frac{\Delta\sigma}{K'}\right)^{1/n'}, \quad (11)$$

where the plastic term is fit in linearized form

$$\frac{\Delta\sigma}{2} = K' \left(\frac{\Delta\varepsilon_p}{2}\right)^{n'} \iff \log\left(\frac{\Delta\sigma}{2}\right) = n' \log\left(\frac{\Delta\varepsilon_p}{2}\right) + \log K', \quad (12)$$

giving the cyclic strength coefficient K' and cyclic strain-hardening exponent n' .

6 Consistency Relations

When the fitted constants are mutually consistent, the following hold and serve as validity checks:

$$n' \approx \frac{b}{c}, \quad K' \approx \frac{\sigma'_f}{(\varepsilon'_f)^{b/c}}. \quad (13)$$

7 Symbol Reference

Symbol	Quantity	Unit
σ, ε	true stress, true strain	Pa, -
$\Delta\varepsilon_t/2$	total strain amplitude	-
$\Delta\varepsilon_e/2$	elastic strain amplitude	-
$\Delta\varepsilon_p/2$	plastic strain amplitude	-
$\Delta\sigma/2$	stress amplitude	Pa
σ_m	mean stress (Morrow)	Pa
R	strain ratio ($\varepsilon_{\min}/\varepsilon_{\max}$, -1 fully reversed)	-
R_{TC}	tension/compression peak-stress ratio	-
$2N_f$	reversals to failure	-
E	Young's modulus	Pa
W	cyclic energy density	MJ m^{-3}
σ'_f, b	fatigue strength coefficient, exponent	Pa, -
ε'_f, c	fatigue ductility coefficient, exponent	-, -
K', n'	cyclic strength coefficient, hardening exponent	Pa, -